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# ForTIFAI: fending off recursive training induced failure for AI model collapse

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The increasing reliance on generative AI models is rapidly increasing the volume of synthetic data, with projections suggesting that most available new data for training could be machine-generated by 2030[1]. This shift to a mainly synthetic data presents a critical challenge: repeated training in synthetic data leads to a phenomenon known as model collapse, where model performance degrades over generations of training, eventually rendering the models ineffective. While the causes of model collapse are increasingly understood, effective mitigation strategies remain scarce. We address this challenge by leveraging a key insight: auto-regressive models tend to generate text sequences to which they assign high confidence. Based on this observation, we introduce Confidence-Aware Loss (CAL) functions, proposing Truncated Cross-Entropy (TCE) as a key instance. CAL mitigates model collapse by down-weighing high-confidence tokens during training, effectively filtering out likely machine-generated artifacts from the learning process. Our experiments demonstrate that models trained with CAL not only learn effectively, but also exhibit significantly increased resilience, tolerating over 2.3 × more synthetic data before the onset of collapse. In addition, we provide an open-source benchmark for collapse dynamics in mixed-data settings. Our results demonstrate that confidence-aware training objectives can substantially delay collapse onset, offering a practical tool for model robustness under synthetic-data exposure.

Generative models have become the foundation for modern AI applications in several modalities, including text, image, code, and audio. Large language models (LLMs) such as ChatGPT[2], LLaMA[3] and Gemma[4], as well as image generators DALL-E[5] and Imagen[6], all rely on large datasets scraped from the Web. As these models are continuously updated to reflect recent knowledge and linguistic patterns, the need for ever larger and frequently refreshed training corpora has grown substantially. However, this demand is colliding with a shift in the data landscape: synthetic content is increasingly populating the Internet, contaminating the very datasets used for model training.

This shift raises fundamental concerns. Generative models trained on outputs from earlier generations can degrade over time, a phenomenon known as *model collapse*[7–9]. Even moderate levels of synthetic data (as little as 1%) have been shown to cause collapse, and scaling the model or the size of the dataset fails to reliably prevent it[10]. Importantly, this is not a hypothetical scenario: recent analysis of Alemohammad et al.[11] on the LAION-5B dataset, a cornerstone for many generative models, revealed measurable contamination from synthetic sources. Following prior works[7,8], we use the term *contamination* or *poisoning* to describe the recursive use of synthetic data in training, which leads to degradation in performance over generations. As synthetic data becomes dominant, safeguarding generative model stability becomes an urgent challenge.

Several approaches have been explored to mitigate model collapse. Some studies combine synthetic and real data to slow degradation[12]; others rely on post hoc supervision, such as SIMS[13], using labeled synthetic data as negative signal to improve generations or explore decoding strategies[14]. In practice, however, distinguishing real from synthetic data remains a major challenge. Watermarking methods[15,16] have been proposed to help identify model-generated content, but they are not widely adopted. Moreover, recent critiques[17] highlight that many prior studies simulate collapse in unrealistic settings, such as using purely synthetic training data, limiting the applicability of their conclusions. Despite growing interest in this issue, the role of the loss function, central to how models learn, has received little attention in the context of model collapse.

Furthermore, model collapse is amplified by approximation errors and finite sampling effects during training, which cause models to concentrate probability mass on a shrinking subset of the true data distribution over generations[8,9].

We observe that generative models are frequently overconfident in their synthetic predictions (Fig. 1). This suggests that abnormally high

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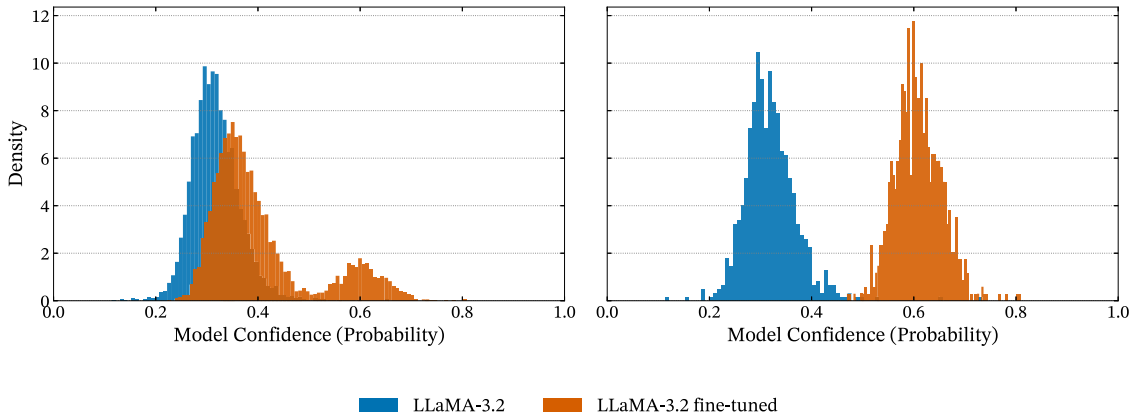



**Fig. 1 | A model exhibits significantly higher confidence in its own generated data compared to unseen data.** The histograms show token probabilities from a base LLaMA-3.2-1B model and a fine-tuned version. *Left*: Confidence distribution when evaluating tokens from WikiText. *Right*: Confidence distribution when evaluating tokens from regenerated WikiText produced by the model. The fine-tuned model shows a clear shift toward higher confidence when evaluating its own synthetic generations, whereas the base model's confidence distribution remains consistent across both datasets.

confidence scores can function as a reliable signal for identifying self-generated data. Building on this observation, we propose confidence-aware loss (CAL) functions, showing that model collapse can be mitigated through these principled adjustments to the training loss. As a primary instance of a CAL function, we introduce *Truncated Cross-Entropy (TCE)*, a loss function that formalizes an intuition of ignoring what the model already "knows" too well. TCE operates by masking any token whose predicted probability exceeds a threshold, effectively preventing the model from overfitting to its own high-confidence, low-diversity outputs. As a minimal extension of the standard cross-entropy (CE) objective, TCE is model-agnostic, computationally efficient, and highly practical for real-world deployment. Furthermore, we identify *Focal Loss (FL)*[18] as another effective CAL variant; our evaluations demonstrate that substituting CE with FL also successfully delays model collapse.

In brief, the contributions of ForTIFAI are as follows:

- We establish that a model's confidence, measured by sequence probability, serves as a strong signal for identifying self-generated text within mixed datasets.
- We introduce CAL functions, specifically proposing TCE and identifying Focal Loss (FL) as key instances. By leveraging this signal to mitigate model collapse, these approaches significantly improve resilience, with TCE enabling models to tolerate more than 2.3 × greater proportions of synthetic data without degradation.
- We propose a comprehensive evaluation framework for model collapse that reflects realistic conditions, including the continual accumulation of new data and the need to measure both retained and newly acquired knowledge.

## Results

To demonstrate the effectiveness of CAL, we primarily evaluate it in the context of large-scale Transformer-based language models. Specifically, we test LLaMA 3[3] and Gemma 3[4] on benchmarks covering mathematical reasoning, logical inference, and factual recall. All Transformer experiments are conducted in full-parameter training settings, demonstrating that CAL mitigates model collapse not only during fine-tuning but also under pre-training-scale optimization. To illustrate the broader applicability of CAL beyond language modeling, To illustrate the broader applicability of CAL beyond language modeling, Supplementary Information sections "Gaussian Mixture Models" and "Variational Auto Encoders" extend the framework to mixture modeling and image generation. In these settings, we apply CAL to VAEs and GMMs, respectively. Across all experimental domains, CAL consistently mitigates collapse while preserving overall learning performance.

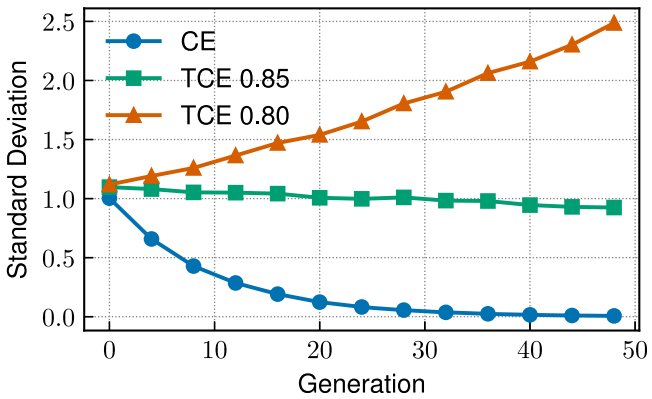



**Fig. 2 | Effect of TCE on a one-dimensional Gaussian estimator under a fully synthetic training loop.** In each generation, 10, 000 samples are generated with sampling bias $\gamma = 0.9$. Standard estimation with Cross Entropy (CE) leads to rapid variance collapse. By contrast, TCE with a well-chosen threshold ($\gamma = 0.85$) delays collapse substantially. Conversely, if the threshold is set too small ($\gamma = 0.80$), the variance may diverge.

### Datasets

We apply the recursive training setup illustrated in Fig. 2 on two datasets: Wikitext-2-raw-v1[19] and a newly curated dataset we call *Imagination of Web*.

Wikitext-2-raw-v1 is a language modeling dataset derived from high-quality Wikipedia articles. Unlike its tokenized counterpart, it preserves raw formatting—including punctuation, casing, and whitespace—making it well-suited for evaluating a model's ability to handle long-range dependencies and natural language structure.

The second dataset, *Imagination of Web*, is a composite dataset constructed from subsets of Wikitext, HellaSwag[20], and GSM8k[21]. It is designed to test the generalization of our loss functions across multiple tasks, such as commonsense, math, and factual recall. This dataset is designed with the goal of being more representative of the common data found on the web.

### Framework parameters

We used the framework described in the Section Model Collapse Evaluation Framework to simulate a progressive model collapse scenario.

For our experiments, we set the number of stages to $N = 6$. We found that model collapse was not reliably observable for fewer than six stages, while increasing $N$ beyond six introduced significant computational overhead with diminishing returns under our resource constraints. For simplicity, we used a fixed amount of data at each stage from the original dataset.

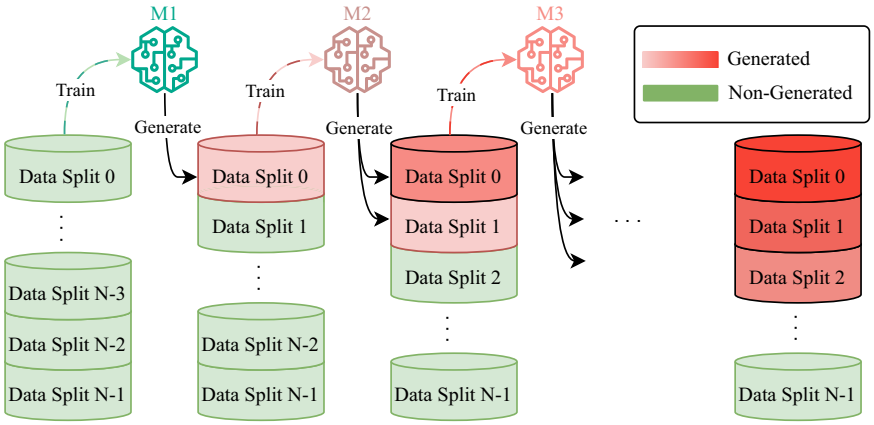



**Fig. 3 | Our experimental setup simulates model collapse, illustrating the transition from predominantly human-generated content to mostly synthetic datasets.** $M_i$ denotes the $i$-th generation of recursively trained models. The clean data is initially split into $N$ equally sized splits. At each generation, the entire dataset from the previous iteration is regenerated and used as training data for the next iteration. One split of clean data is added to this training dataset to simulate the accumulation of non-generated data into the training set. The red color for each data split indicates the contamination from recursively training on the self-generated data, resulting in lower quality (darker red color).

### Hyperparameters

For both LLaMA and Gemma model families, we tuned and used the best-performing hyperparameters. All models were fine-tuned using a learning rate of $2 \times 10^{-5}$, with a batch size of 64. The training was conducted over 10 epochs per stage with the best performing checkpoints saved. In order to have a fair comparison between different loss functions, we ran several identical experiments with different $\gamma$ values for both Focal and TCE losses as shown in Table 5 for the LLaMA-3.2-1B model and fix these parameters for the Gemma-3-1b-pt model as well.

### Model scale considerations

All experiments are conducted in a full-parameter training setting rather than parameter-efficient fine-tuning (e.g., LoRA[22]). This design choice is deliberate: model collapse is a distributional degeneration phenomenon that emerges through iterative retraining, and partial parameter updates may not faithfully reproduce the dynamics observed under full-model optimization.

Due to the substantial computational cost of repeated multi-stage full-parameter retraining, our largest evaluated Transformer models contain 1.5B parameters. Extending the framework to larger scales would require significantly greater computational budgets.

Intuitively, larger models better approximate the underlying data distribution, reducing early-stage over-amplification of high-confidence tokens. We therefore expect the qualitative behavior of CAL to persist at larger scales, though comprehensive large-scale validation remains important future work. Further details are provided in the Supplementary Information section "Inevitable Model Collapse".

### Knowledge retention test

To comprehensively assess the impact of our loss functions on model performance and resilience to model collapse, we perform evaluations on a mix of standard and custom benchmarks. These benchmarks test various capabilities, including factual recall, linguistic coherence, commonsense, and mathematical reasoning. In addition to widely used benchmarks like BLiMP, Hellaswag, and GSM8K, we utilize a question-answering benchmark—*Knowledge Retention test* (KR-test)[23] to measure how well models retain training-set facts.

Validation perplexity is a common metric for evaluating LLMs, but it primarily reflects predictive fluency; how well a model guesses the next token in unseen text. While useful for assessing grammar and style, it does not quantify factual knowledge retention. Moreover, low perplexity does not guarantee meaningful or human-like output, as humans do not always select the most likely next word.

To address this gap, we use the *Knowledge Retention Test* (KR-test)[23], which measures factual *retention* from the training data rather than general fluency.

Given a body of training text, we divide it into complete paragraphs and generate five questions per paragraph. These questions and completions can be crafted manually or using frontier LLM APIs such as ChatGPT[2] or Gemini[4]. Each question consists of a context, a true continuation, and a false continuation.

We compute the log probability of the model for two sequences:

- *Context + Factually correct completion*
- *Context + Factually incorrect completion*

When the log probability is higher for the true continuation, the question is considered correctly answered. The KR-test score is the model's total accuracy over all the questions.

A high KR-test score indicates that the model has internalized the factual content of the paragraph. This benchmark specifically tests factual retention from the training data. More details on KR-Test are provided in the Supplementary Information.

Figure 3 illustrates the total KR-accuracy of the LLaMA-3.2-1B model trained on the WikiText dataset using three distinct loss functions: (CE, the baseline), and two instances of CAL: Focal Loss, and TCE. Across the sequential training stages *(M0–M5)*, the model must acquire new information while retaining knowledge from previous stages. As expected, KR-accuracy trends upward, reflecting the model's growing capacity to answer questions based on the newly introduced data. Crucially, both TCE and Focal Loss demonstrate superior efficacy in learning from fresh data and preserving prior knowledge, consistently yielding higher accuracy across all stages compared to the CE baseline.

An illustrative example of KR-test questions is provided in Table S2, showing how the model must assign higher likelihood to factually correct continuations over incorrect ones. This clarifies the evaluation protocol used to compute factual-retention accuracy. Statistics regarding the dataset itself are also provided in Table S1 in terms of the length of the contexts and completions. The distribution of the completions' difficulty is also provided in Supplementary Information Knowledge Retention Test (KR-Test).

### Math, reasoning, and coherence

In addition to factual retention, we evaluate model's performance across three reasoning and language understanding benchmarks. These evaluations assess whether our loss functions preserve or enhance generalization and linguistic structure, both of which are vulnerable under recursive training regimes.

To evaluate the impact of model collapse on reasoning and linguistic capability, we use three standard benchmarks: GSM8K [21], Hellaswag [20], and BLiMP [24].

- *GSM8K* measures mathematical reasoning by testing the model's ability to solve grade-school level word problems. Success on this benchmark reflects the model's ability to perform multi-step arithmetic reasoning.

- *Hellaswag* evaluates commonsense reasoning through multiple-choice sentence completion tasks. It requires the model to infer plausible continuations based on everyday scenarios.
- *BLiMP* tests grammatical coherence and syntactic structure. It consists of minimal pairs where the model must choose the grammatically correct option. High performance indicates strong control over syntax and language structure.

Together, these benchmarks assess the model's ability to generalize, reason, and maintain coherence under different forms of linguistic stress.

**Table 1 | Comparison of LLaMA and Gemma models trained on M0 (clean, non-synthetic data) from the Wikitext dataset.**

| Model | Method | Blimp | Hellaswag | KR-test | Average |
|---|---|---|---|---|---|
| LLaMA | Baseline | 75.1% | 58.5% | 64.0% | 65.86% |
| | Focal ($\gamma = 2$) | 75.5% | **62.5%** | **64.9%** | 67.63 |
| | TCE ($\gamma = 0.9$) | **76.8%** | 61.8% | 64.8% | **67.8%** |
| Gemma | Baseline | 71.2% | 63.9% | 67.7% | 67.6% |
| | Focal ($\gamma = 2$) | 73.5% | **64.9%** | 67.4% | 68.6 |
| | TCE ($\gamma = 0.9$) | **73.9%** | 64.6% | **68.2%** | **68.9%** |

Models were evaluated across the Blimp, Hellaswag, GSM8k, and KR-test benchmarks. For both architectures, TCE and Focal loss outperform the baseline (CE) when trained on real data The highest value per block is bolded.

**Table 2 | Comparison of M5 benchmark results across LLaMA and Gemma models for Baseline, TCE ($\gamma = 0.9$), and Focal ($\gamma = 2$) on *Wikitext dataset***

| Model | Method | Blimp | Hellaswag | KR-test | Average |
|---|---|---|---|---|---|
| LLaMA | Baseline | 63.3% | 45.7% | 69.2% | 59.4% |
| | Focal ($\gamma = 2$) | **67.5%** | **49.6%** | **72.4%** | **63.16** |
| | TCE ($\gamma = 0.9$) | 63.6% | 48.4% | 71.4% | 61.13% |
| Gemma | Baseline | 61.2% | 52.4% | 68.1% | 60.56% |
| | Focal ($\gamma = 2$) | **64.8%** | 51.6% | 69.2% | **61.86** |
| | TCE ($\gamma = 0.9$) | 61.8% | **53%** | **69.7%** | 61.5% |

The highest value per block is bolded. While TCE and Focal consistently surpass the baseline, neither shows a clear advantage over the other, indicating that with a well-tuned $\gamma$ value, CAL functions tend to have similar performance.

For GSM8K and Hellaswag, we use the Language Model Evaluation Harness[25] to ensure consistent evaluation across models.

### Wikitext dataset

To assess performance under idealized, non-collapsing conditions, we also evaluate models trained exclusively on original, non-synthetic data. The results, shown in Table 1, confirm that CAL on both instances TCE ($\gamma = 0.9$) and Focal ($\gamma = 2$) maintain strong performance in streamline training settings. Ultimately, these findings demonstrate that CAL consistently outperforms standard CE, even outside of model collapse scenarios.

Table 2 presents the results after 5 stages of synthetic data accumulation as discussed in Section Model Collapse Evaluation Framework. The evaluation spans across the full benchmark suite, excluding GSM, since the Wikitext training dataset does not include any math or reasoning. Overall, TCE and Focal loss outperform CE in preserving factual knowledge. While TCE and Focal consistently surpass the baseline, neither shows a clear advantage over the other, indicating that with a well-tuned $\gamma$ value, CAL functions perform similarly.

Performance of both models when the experiment is conducted on Wikitext dataset is shown in Fig. 4, where both CAL variants consistently outperform CE during different stages of the experiment.

### Imagination-of-Web dataset

To evaluate in a setting closer to real-world pre-training, we adopt the *Imagination-of-Web* (IoW) corpus—a composite of factual prose (Wikitext), commonsense reasoning (HellaSwag), and mathematical problem solving (GSM8K).

As detailed in Table 3, we first establish the baseline performance using only the pristine, human-generated IoW data. Consistent with our earlier findings, both CAL variants (TCE ($\gamma = 0.9$) and FL ($\gamma = 2$)) exhibit robust performance in this standard training regime. This reinforces the conclusion that CAL provides a distinct advantage over the conventional CE approach, even in the absence of data degradation.

Turning to the iterative training scenario, Table 4 illustrates the outcomes following 5 generations of synthetic data accumulation. Because the IoW corpus encompasses a broader range of domains, this evaluation successfully covers our entire benchmark suite, including the math and reasoning tasks (GSM) that were previously omitted. Across this comprehensive suite, both TCE and Focal loss mitigate performance decay significantly better than standard CE. Furthermore, the parallel success of TCE and Focal loss across these diverse benchmarks reinforces our prior observation: when the scaling factor $\gamma$ is appropriately calibrated, the specific choice of CAL formulation is largely inconsequential, as both deliver comparable and highly effective resistance to model collapse.

Performance of both models when the experiment is conducted on the Imagination dataset is shown in Fig. 4, where CAL consistently outperform CE during different stages of the experiment.

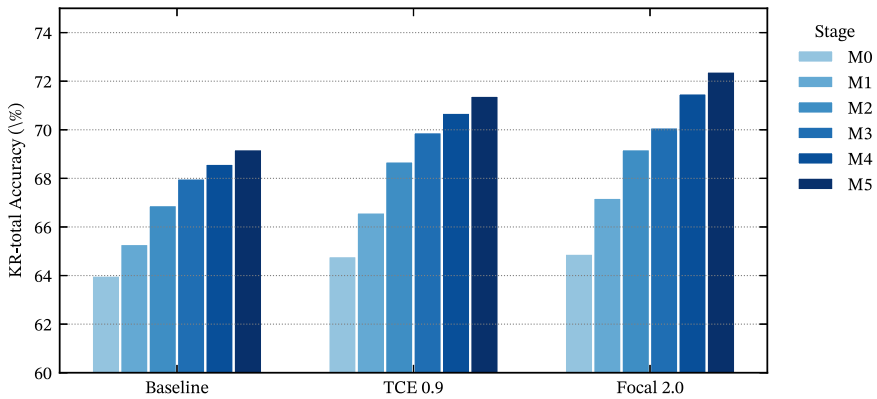



**Fig. 4 | Total knowledge retention (KR-total) accuracy with LLaMA-3.2-1B model trained on Wikitext.** TCE and Focal loss retain learning capability and exceed the baseline throughout different stages.

**Table 3 | Comparison of LLaMA and Gemma models on M0 (clean and non-generated data) across Blimp, Hellaswag, GSM8k, and KR-total on *Imagination dataset***

| Model | Method | Blimp | Hellaswag | GSM8k | KR-total | Average |
|---|---|---|---|---|---|---|
| LLaMA | Baseline | 76.3% | 63.1% | 8.1% | 63.6% | 52.77% |
| | Focal ($\gamma = 2$) | **79.6%** | **65.2%** | **9.5%** | **64.8%** | **54.77%** |
| | TCE ($\gamma = 0.9$) | 76.8% | 64.6% | 9.1% | 64.8% | 53.82% |
| Gemma | Baseline | 73.1% | 65% | **6.2%** | 64.1% | 52.1% |
| | Focal ($\gamma = 2$) | **77.2%** | **65.7%** | 4.5% | 64.2% | **52.9%** |
| | TCE ($\gamma = 0.9$) | 76% | 65% | 5.2% | **64%** | 52.55% |

The highest value per block is bolded, and the second highest is underlined.

**Table 4 | Comparison of M5 benchmark results across LLaMA and Gemma models for Baseline, TCE ($\gamma = 0.9$), and Focal ($\gamma = 2$) on *Imagination dataset***

| Model | Method | Blimp | Hellaswag | GSM8k | KR-total | Average |
|---|---|---|---|---|---|---|
| LLaMA | Baseline | 72.1% | 52.0% | 5.9% | 68.7% | 49.67 |
| | Focal ($\gamma = 2$) | 76.4% | 51.1% | 3.1% | **72.7%** | 50.82 |
| | TCE ($\gamma = 0.9$) | **77.6%** | **54.2%** | **6.2%** | 71.0% | **55.25%** |
| Gemma | Baseline | 71% | 53.4% | 3.9% | 71.6% | 49.97% |
| | Focal ($\gamma = 2$) | **74.2%** | **55%** | 4% | 70.9% | **51.02** |
| | TCE ($\gamma = 0.9$) | 71.7% | 53.1% | **4.2%** | **74.1%** | 50.77% |

The highest value per block is bolded.

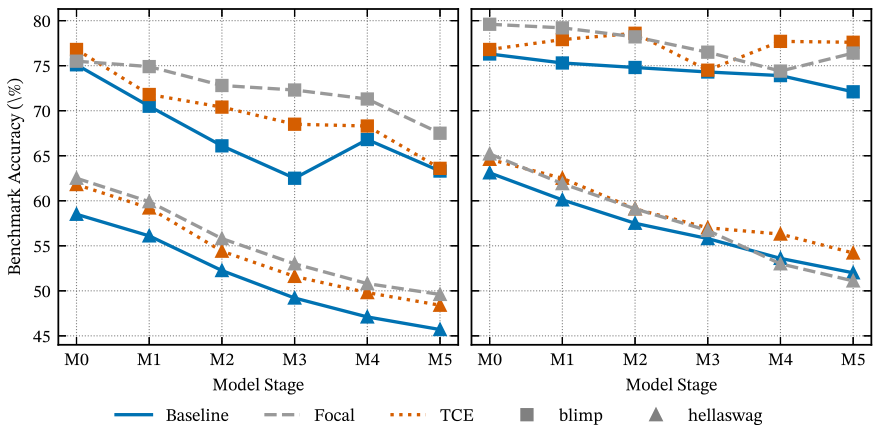



**Fig. 5 | Our proposed solution outperforms the CE baseline consistently across all benchmarks.** *Left*: Wikitext training. *Right*: Imagination training.

## Time to failure

We define *time to failure* as the number of re-training tokens it takes a model in the context of this framework to degrade significantly compared to the initial model. Specifically, we analyze the performance of the model on the first partition of the dataset (KR-test consisting only of questions from M0), which is re-generated at every stage. Moreover, since the KR-test on average has a baseline of 50% and a maximum of 100% is achievable, we define failure as performance dropping below 75%, which is exactly half the maximum gain one can expect from training the model. As shown in Fig. 5, CE fails between the first and second stages, while both CAL instances extend time to failure by over 2.3 × . These results suggest CAL is effective at sustaining meaningful retention under compounding self-generation.

## KL divergence

To quantify distributional drift across generations, we measured the Kullback-Leibler (KL) divergence between the token distributions of synthetic datasets and the original WikiText-2 corpus. Each dataset is tokenized with the model's tokenizer, and normalized token frequencies are used to compute the KL divergence between the original and synthetic distributions.

To evaluate distributional fidelity, we tracked the KL divergence between generated outputs and the original training distribution. Figure 6 demonstrates that CE leads to rapid divergence, indicative of distributional drift and collapse. On the other hand, both FL and TCE demonstrate less divergence from the original distribution.

## Ablation study

We investigate the impact of the confidence parameter $\gamma$ in both TCE and Focal Loss on learning performance and robustness to model collapse. The parameter $\gamma$ controls the strength of confidence suppression: overly aggressive settings may discard useful gradient signal, while overly permissive settings reduce the method toward standard CE. To provide

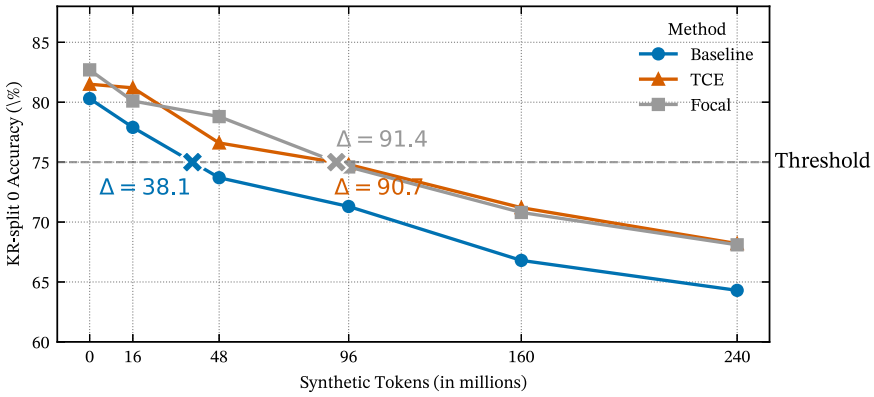



**Fig. 6 | Our proposed loss functions effectively mitigate model collapse in a fully synthetic subset of the dataset.** The figure above corresponds to LLaMA-3.2-1B's accuracies on answering questions related to the dataset they were trained on. Red dotted lines provided for better comparison of time to failure. The number of stages until failure is shown as Δ. Time-to-failure comparison of loss functions on synthetic subsets of WikiText. Dotted red lines indicate failure threshold (75% KR accuracy). Both CAL instances outperform baseline with similar gaps.

**Table 5 | Time-to-failure (TTF) across loss functions and $\gamma$ values under recursive retraining.**

| Method | Time-to-Failure (Million Tokens) ↑ |
|---|---|
| Baseline (CE) | 38.10 |
| TCE ($\gamma$ = 0.8) | 39.89 |
| TCE ($\gamma$ = 0.85) | 33.97 |
| **TCE ($\gamma$ = 0.9)** | **90.67** |
| TCE ($\gamma$ = 0.99) | 68.09 |
| Focal ($\gamma$ = 1) | 18.67 |
| **Focal ($\gamma$ = 2)** | **91.43** |
| Focal ($\gamma$ = 4) | 31.60 |

practitioners with actionable guidance, Table 5 reports the effect of varying $\gamma$ across training stages (M0-M5) for the LLaMA-3.2-1B model on WikiText.

Across all stages, every tested $\gamma$ for both TCE and FL improves upon CE, indicating that CAL methods are not highly sensitive to precise hyperparameter selection. However, a consistent pattern emerges: *moderate* values of $\gamma$ provide the most favorable trade-off between collapse mitigation and preservation of general capabilities.

For TCE, values in the range $\gamma \in [0.85, 0.9]$ repeatedly achieve top or near-top performance. For focal loss, $\gamma = 2$ provides the most stable improvements across benchmarks, consistent with prior findings[26], while larger values can overly suppress gradients and degrade later-stage performance.

Intuitively, small $\gamma$ in TCE aggressively truncates high-confidence tokens, potentially removing informative signal. Similarly, large $\gamma$ in FL excessively re-weights gradients. Moderate values selectively suppress the extreme high-confidence tail, where recursive collapse manifests, without destabilizing optimization.

In practice, we recommend inspecting confidence histograms and selecting $\gamma$ to trim only the extreme high-confidence region while retaining the bulk of the distribution. Our experiments suggest $\gamma \approx 0.85$-$0.9$ for TCE and $\gamma \approx 2$ for FL as robust starting points. A deeper theoretical characterization of optimal $\gamma$ selection across domains remains an important direction for future work.

Table 6 reveals how the time-to-failure changes based on $\gamma$ values. Notably, TCE with $\gamma = 0.9$ and FL with $\gamma = 2$ increase time-to-failure by more than 2 × compared to CE, while neighboring values degrade performance. We emphasize that time-to-failure is used strictly as a diagnostic metric for collapse dynamics on the most vulnerable split and does not, by itself, capture performance on unaffected splits or external benchmarks.

Complete numerical results across all benchmarks and datasets are reported in Tables S3–S5. These tables provide full per-generation accuracies for both LLaMA and Gemma on the WikiText and Imagination-of-Web corpora, respectively. The detailed results corroborate the main-text findings, confirming that confidence-aware objectives consistently outperform standard Cross Entropy throughout recursive training.

## Discussion

We have demonstrated that a CAL function, namely TCE, and FL, provide a simple and powerful mitigation strategy against model collapse. Grounded in a mathematical framework linking overconfidence to distributional drift, CAL extends a model's operational lifetime by over 2.3 × compared to standard CE. Our benchmarks confirm that this approach not only delays collapse but also improves knowledge retention and generalization in environments with accumulating synthetic data.

Crucially, the principle of filtering overconfident outputs is not domain-specific. We verified its efficacy in diverse generative systems, including Gaussian Mixture Models and Variational Autoencoders, as shown in the Supplementary Information. As the line between human and machine-generated data continues to blur, such robust, low-overhead training methods will be essential for building stable and trustworthy AI.

Beyond language modeling, Fig. S2 and Algorithm S1 show that applying the truncation strategy to Gaussian Mixture Models effectively delays distributional collapse, while Fig. S3 demonstrates that a truncated VAE maintains image fidelity across recursive self-generation stages. These experiments confirm that the proposed CAL generalizes to a wide range of generative architectures.

We approached model collapse from a training perspective and proposed a class of loss functions that mitigate collapse effectively without sacrificing performance. By reweighting token probabilities to favor underrepresented or underestimated outputs, these losses retain the simplicity of CE while offering greater resilience during recursive training.

Looking ahead, several research avenues merit further exploration:

- *Broader model classes*: Model collapse has been extensively studied in image generation domains. It remains an open question whether similar loss-based strategies can be adapted to diffusion models[27] and generative adversarial networks (GANs)[28], where diversity collapse is also a central concern.
- *Heterogeneous model ecosystems*: Our setup assumes a single model trained recursively across generations. However, real-world content generation is inherently multi-agent and heterogeneous, involving a mixture of open and closed-source models. Extending our analysis to simulate ecosystems of interacting generative models could provide deeper insights into how synthetic content evolves when shaped by a plurality of architectures and training paradigms.
- *Collapse-aware generation*: While our approach targets the training process, model collapse can also be addressed during the generation

**Table 6 | LLaMA-3.2-1B unified benchmark results across Blimp, KR-split$_0$, and KR-total of trainings on Wikitext to set the values of $\gamma$ for each loss function across different models and datasets using LLaMA-3.2-1B.**

| Stage | Method | Blimp | Hellaswag | GSM8k | KR - split$_0$ | KR-total |
|---|---|---|---|---|---|---|
| M0 | Baseline | 75.1% | 58.5% | 2.5% | 80.3% | 64.0% |
| | Focal ($\gamma$ = 1) | 75.29% | 61.6% | 2.7% | 81.38% | 65.62% |
| | Focal ($\gamma$ = 2) | 75.5% | 62.5% | 2.8% | 82.7% | 64.9% |
| | Focal ($\gamma$ = 4) | 76.08% | **63.8%** | 2.2% | 82.35% | **66.98%** |
| | TCE ($\gamma$ = 0.8) | 75.59 | 62% | 2.3% | 81.85% | 66.73% |
| | TCE ($\gamma$ = 0.85) | 76.64 | 61.6% | 2.5% | **83.09%** | 66.52% |
| | TCE ($\gamma$ = 0.9) | **76.8%** | 61.8% | **2.9%** | 81.5% | 64.8% |
| | TCE ($\gamma$ = 0.99) | 75.5% | 59.9% | 2.5% | 81.7% | 64.0% |
| M1 | Baseline | 70.5% | 56.1% | 1.8% | 77.9% | 65.3% |
| | Focal ($\gamma$ = 1) | 72.17% | 58.7% | 2% | 75.31% | 70.76% |
| | Focal ($\gamma$ = 2) | 74.9% | 59.9% | 2.1% | 80.1% | 67.2% |
| | Focal ($\gamma$ = 4) | **75.34%** | **60%** | 1.7% | 78.93% | 72.69% |
| | TCE ($\gamma$ = 0.8) | 74.87% | 59.0% | 1.3% | 80.69% | 73.08% |
| | TCE ($\gamma$ = 0.85) | 74.42% | 59.2% | **2.3%** | 79.13% | **73.32%** |
| | TCE ($\gamma$ = 0.9) | 71.8% | 59.2% | 2.2% | **81.2%** | 66.6% |
| | TCE ($\gamma$ = 0.99) | 73.3% | 57.1% | 1.2% | 78.5% | 65.1% |
| M2 | Baseline | 66.1% | 52.25% | **1.8%** | 73.7% | 66.9% |
| | Focal ($\gamma$ = 1) | 70.42% | 55.7% | 0.9% | 71.75% | 75.36% |
| | Focal ($\gamma$ = 2) | **72.8%** | 55.8% | 1.6% | **78.8%** | 69.2% |
| | Focal ($\gamma$ = 4) | 71.79% | **57.3%** | **1.8%** | 70.9% | 76.35% |
| | TCE ($\gamma$ = 0.8) | 70.89% | 56.2% | 1.7% | 73.17% | **76.51%** |
| | TCE ($\gamma$ = 0.85) | 71.17% | 56.3% | 1.4% | 71.83% | 76.11% |
| | TCE ($\gamma$ = 0.9) | 70.4% | 54.4% | 1.2% | 76.6% | 68.7% |
| | TCE ($\gamma$ = 0.99) | 67.1% | 53.4% | 1.5% | 76.8% | 67.6% |
| M3 | Baseline | 62.5% | 53% | 1.1% | 71.3% | 68.0% |
| | Focal ($\gamma$ = 1) | 67.14% | 53% | **1.5%** | 69.1% | 71.38% |
| | Focal ($\gamma$ = 2) | **72.3%** | 53.0% | 1.1% | 74.6% | 70.1% |
| | Focal ($\gamma$ = 4) | 72.22% | **54.86%** | 1.2% | 69.98% | **73.56%** |
| | TCE ($\gamma$ = 0.8) | 64.76% | 53.2% | 1.1% | 73.17% | 72.83% |
| | TCE ($\gamma$ = 0.85) | 71.18% | 53.3% | 1.2% | 74.47% | 73.19% |
| | TCE ($\gamma$ = 0.9) | 68.5% | 51.6% | 0.9% | **74.8%** | 69.9% |
| | TCE ($\gamma$ = 0.99) | 66.8% | 50.6% | 1.2% | 72.5% | 69.9% |
| M4 | Baseline | 66.8% | 47.1% | 0.7% | 66.8% | 68.6% |
| | Focal ($\gamma$ = 1) | 69.35% | 51.9% | 0.8% | 66.86% | 67.91% |
| | Focal ($\gamma$ = 2) | 71.3% | 50.8% | 0.6% | 70.8% | **71.5%** |
| | Focal ($\gamma$ = 4) | 71.03% | 52.3% | 1.6% | 66.79% | 70.48% |
| | TCE ($\gamma$ = 0.8) | 69.68% | **52.5%** | **2.0%** | 69.72% | 70.06% |
| | TCE ($\gamma$ = 0.85) | **71.38%** | 51.5% | **2.0%** | 70.31% | 69.1% |
| | TCE ($\gamma$ = 0.9) | 68.3% | 49.8% | 0.8% | **71.2%** | 70.7% |
| | TCE ($\gamma$ = 0.99) | 69.9% | 48.3% | 0.7% | 69.1% | 70.4% |
| M5 | Baseline | 63.3% | 45.7% | 1% | 64.3% | 69.2% |
| | Focal ($\gamma$ = 1) | 66.06% | 51.2% | 1.2% | 66.72% | 67.52% |
| | Focal ($\gamma$ = 2) | 67.5% | 49.6% | 1.1% | 68.1% | **72.4%** |
| | Focal ($\gamma$ = 4) | **70.45%** | **52.2%** | 1.5% | 67.87% | 68.19% |
| | TCE ($\gamma$ = 0.8) | 70.1% | 49.17% | 1.2% | **70.33%** | 67.71% |
| | TCE ($\gamma$ = 0.85) | 66.99% | 50.6% | **1.6%** | 67.77% | 66.61% |
| | TCE ($\gamma$ = 0.9) | 63.6% | 48.4% | 0.9% | 68.2% | 71.4% |
| | TCE ($\gamma$ = 0.99) | 62.6% | 46.9% | 0.8% | 66.2% | 70.2% |

For ease of readability, we highlight the highest accuracy in bold and the second best as underlined for every stage of the framework.

phase. Combining our training-side solution with decoding-aware techniques, such as[14] may further improve the quality and diversity of generated data.

## Methods

### Problem definition

Model Collapse is the gradual degradation that occurs when a model's own outputs pollute the training data of its successors. Each new model then learns from these possibly flawed examples, causing a drift from the true data distribution[7].

The fundamental cause of collapse is the discrepancy between the true data distribution, $P_{\text{data}}$, and the model's learned approximation of it, $P_\theta$. Because any realizable model has finite capacity and is trained on finite data, its outputs will contain statistical artifacts not present in the true distribution. Compounding this issue, the decoding process itself introduces a further distributional shift. Common decoding algorithms (e.g., greedy search, beam search) are designed to favor high-probability sequences rather than sampling purely from the model's learned distribution. This bias means that even if a model were a perfect representation of the data ($P_\theta = P_{\text{data}}$), the generated text would over-represent the modes of the distribution.

When these skewed outputs are used to train a successor model, the imperfections are learned and amplified. This iterative process leads to a progressive distributional drift, causing a catastrophic loss of information, particularly in the tails of the distribution, ultimately making the model unusable.

### Key observation

Figure 1 illustrates a consistent gap in model confidence between self-generated samples and unseen data. We exploit this difference as a signal to identify and remove the effect of synthetic samples during training. This gap is a side effect of the approximation error and the greedy sampling performed for generating text [14].

### Truncated cross entropy (TCE)

Building on the findings from the Key Observation section, we introduce a novel loss function, TCE. In TCE, confident predictions are explicitly masked to remove their influence on training. By selectively masking overly confident predictions, TCE forces the model to learn from less-certain datapoints, thereby preserving distribution tails compared to CE. We formulate TCE as:

$$TCE(p_t) = \chi_\gamma(p_t) \times CE(p_t) \tag{1}$$

$$\chi_\gamma(p_t) = \begin{cases} 1 \; if p_t \le \gamma \\ 0 \; if p_t > \gamma \end{cases} \tag{2}$$

Here, $p_t$ is the probability of the correct class, $\chi_\gamma(.)$ is an indicator function, $CE(.)$ is the CE loss, and $\gamma \in [0, 1]$ is the confidence threshold. Setting $\gamma$ close to 1 recovers standard CE loss, while smaller values selectively suppress high-confidence predictions. Due to its explicit truncation mechanism and minimal modification to CE, TCE provides a clean and interpretable instantiation of CAL functions, enabling direct analysis of how suppressing high-confidence tokens affects model collapse dynamics. In the Mathematical Insight section, we further substantiate this intuition with a theoretical analysis of TCE.

A natural concern is that TCE might degrade foundational knowledge by masking high-confidence tokens (where probability exceeds $\gamma$), preventing their reinforcement. However, this truncation is conditional. If a token's confidence drops below $\gamma$ due to distributional shifts or retraining noise, it automatically re-enters the training signal. Thus, TCE acts as an adaptive regulator, allowing foundational knowledge to self-stabilize near the threshold and receive reinforcement only when certainty weakens. Empirically (as shown in the "Results" section), TCE preserves core capabilities without degrading standard benchmark performance.

### Focal loss

FL, originally introduced for addressing class imbalance in image segmentation tasks [18], emphasizes hard, misclassified examples. It is defined as:

$$FL(p_t) = -(1 - p_t)^\gamma \times \log(p_t) \tag{3}$$

where $p_t$ is the probability of the correct class, and $\gamma$ is a tunable hyperparameter that controls the down-weighting of well-classified samples. FL reduces the loss for confident predictions, acting similarly to CE on uncertain examples, thus helping avoid overfitting to self-generated tokens.

Recent work [26] has shown FL's effectiveness in language modeling, outperforming CE on commonsense reasoning and math tasks. In the context of this work, FL aligns with our objective of mitigating model collapse by emphasizing rare token learning and preserving the tail of the output distribution.

### Mathematical insight

The growing reliance on synthetic data in generative model training pipelines has given rise to a self-consuming feedback loop wherein models are recursively trained on their own outputs.

In a fully synthetic training loop, where each model $G_t$ is trained solely on samples from $G_{t-1}$, [11] prove that the model's estimated covariance $\sigma_t$ collapses almost surely:

$$\mathbb{E}[\sigma_t^2 | \sigma_{t-1}^2] = \lambda \sigma_{t-1}^2, \;\; \text{with} \, \lambda \le 1, \Rightarrow \; \sigma_t^2 \xrightarrow{a.s.} 0. \tag{4}$$

where $\lambda$ represents sampling bias, contracting the variance at each iteration, leading to rapid loss of sample diversity, as illustrated in Fig. 7. This decrease in variance is empirically observed across models such as DDPMs[27] and StyleGAN-2[29].

To build intuition on how to counteract this collapse, we analyze a one-dimensional Gaussian model where data is drawn from $X_t \sim \mathcal{N}(0, \sigma_t^2)$. Our strategy is to modify the training objective to focus on low-probability samples. In our Gaussian case, this is equivalent to training only on samples from the tails of the distribution, i.e., where $|X_t| \ge a\sigma_t$ for a chosen threshold $a > 0$.

The conditional variance of this truncated distribution is given by:

$$\text{Var}(X_t \,|\, |X_t| \ge a\sigma_t) = \eta(a) \cdot \sigma_t^2, \tag{5}$$

where $\eta(a)$ is a variance amplification factor. Using known properties of the truncated normal distribution, this factor is calculated as:

$$\eta(a) = 1 + \frac{a\,\phi(a)}{1 - \Phi(a)} \tag{6}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF, respectively. Crucially, $\eta(a) > 1$ for all $a > 0$.

By incorporating this variance amplification into the self-consuming loop, the recurrence for the variance becomes:

$$\mathbb{E}[\sigma_{t+1}^2] = \lambda \cdot \eta(a) \cdot \mathbb{E}[\sigma_t^2]. \tag{7}$$

The collapse can now be slowed by choosing a threshold $a$ such that the combined factor $\lambda \cdot \eta(a) \approx 1$, stabilizing the training process by preserving sample diversity.

We argue that overconfident predictions on common tokens can dominate the training signal, ultimately causing the model to underrepresent less frequent but meaningful patterns. If not addressed, this effect progressively narrows the model's output distribution, highlighting the importance of suppressing overconfident learning.

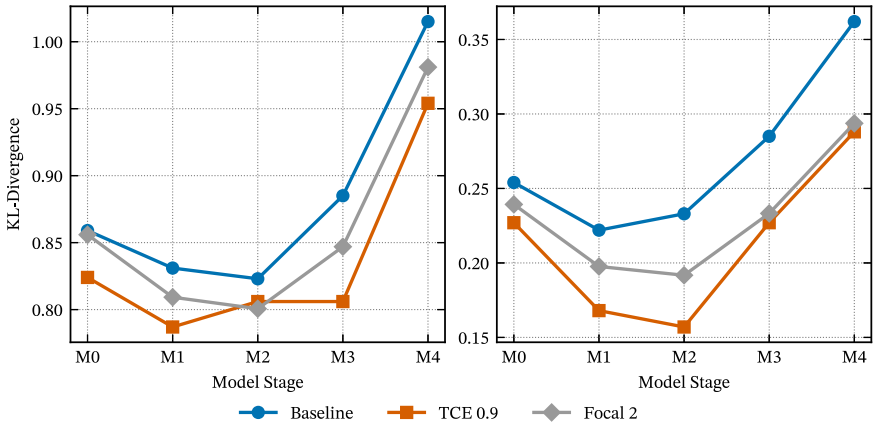



**Fig. 7 | To quantify collapse dynamics, we measured the Kullback-Leibler (KL) divergence between the distribution of model-generated outputs and the original training data distribution over successive generations of the WikiText dataset. Left:** Results for LLaMA-3.2-1B. **Right:** Results for Gemma. Models trained with Cross Entropy show rapidly increasing KL divergence, indicating distributional drift and loss of diversity. In contrast, CAL functions significantly slow the divergence growth, effectively preserving distributional similarity across generations.

Both Focal Loss (FL) and TCE aim to suppress the dominance of high-confidence predictions; however, they differ fundamentally in how they modulate the training signal. Focal Loss applies a smooth multiplicative reweighting of the gradient across the entire confidence spectrum, scaling each token's contribution by $(1-p)^{\gamma}$, where $p$ is the predicted probability of the ground-truth token. Consequently, even highly confident predictions continue to receive non-zero gradient updates, albeit attenuated. In contrast, TCE implements a hard truncation: tokens whose confidence exceeds a threshold are completely masked from gradient updates. Under the Gaussian model, this is equivalent to conditioning exclusively on the tail region $|X_t| \geq a\sigma_t$, yielding an explicit variance amplification factor $\eta(a) > 1$.

Figure S1 illustrates the one-dimensional Gaussian model-collapse example under both perfect and noisy estimators, demonstrating the progressive disappearance of distribution tails. Further details are presented in Supplementary Information Inevitable Model Collapse. Table S6 further presents examples of text-generation degradation across successive stages, highlighting how approximation errors and recursive self-training compound to accelerate collapse.

### Model collapse evaluation framework

To realistically assess the impact of *Model Collapse*, we adopt an experimental framework inspired by ref. 12. Initially, web data was predominantly human-generated and often curated for quality. Over time, however, machine-generated content, often indistinguishable from human-written text, has increasingly supplemented online data. Our framework simulates a realistic model collapse process by progressively adding synthetic data to the training set. We evaluate the methods on math, reasoning, and knowledge recall benchmark performance. Moreover, we analyze the number of synthetic training iterations it takes for the model to lose performance below a certain threshold, which we call the time to failure.

As illustrated in Fig. 2, the experiment proceeds in multiple stages. At each stage, the dataset simulates a large-scale pretraining corpus, composed of both authentic and self-generated data. At every step, we regenerate the previous stage's dataset using the latest model and concatenate the dataset with a fixed amount of new human-written text. This new dataset is then used to further train the model, as detailed further in Algorithm 1.

**Algorithm 1. Recursive data generation with an LLM**

**Inputs:**
$N \geq 2$
$N+1$ splits of authentic dataset $D_0^{(0)}, \ldots, D_N^{(0)}$, each of equal size
▷ $D_i^{(j)}$ is the $j-$ thgeneration of the $i-$ th split.
$N+1$ evaluation sets $Eval_0, \ldots, Eval_N$, each of equal size
▷ $Eval_i$ is based on the content of $D_i^{(0)}$.
$M_0 \Leftarrow$ Pretrained model
**for** $i = 0 \rightarrow N$ **do**
  Train $M_i$ on $[D_0^{(i)}, D_1^{(i-1)}, \ldots, D_i^{(0)}]$
  **for** $k = 0 \rightarrow i$ **do**
  $D_k^{(i+1-k)} \Leftarrow M_i(D_k^{(i-k)})$
  Evaluate $M_i$ on $Eval_k$
  **end for**
  $M_{i+1} \Leftarrow M_i$
**end for**

This setup also follows the conditions mentioned in [17]:

1. *Increasing pre-training data*: Unlike previous studies with fixed dataset sizes, our setup continuously increases the pre-training data at each generation. This growth mirrors real-world trends among state-of-the-art models; for example, LLaMA 1, 2, and 3 were trained on 1.4, 2, and 15 trillion tokens, respectively. Although dataset growth has eventual limits, these bounds remain extremely high (approximately one quadrillion tokens)[30], far exceeding current dataset sizes.
2. *Synthetic data accumulation alongside real data*: Another distinctive feature of our framework is its simultaneous training on both real and synthetic data. Unlike previous work, we do not discount the effect of real unseen data on model collapse. We argue that the real unseen data is the main reason behind training the new generation of models in the first place.
3. *Decreasing proportion of real data*: A primary concern highlighted in recent literature is the accelerating generation rate of synthetic data and its impact on future generative models. Our experimental setup realistically emulates this crucial aspect (e.g., in our setup with 6 stages, we have 100%, 50%, 33%, 25%, 20%, and 15% real data at each stage, respectively).

## Data availability

All data is available at https://github.com/pedramaghazadeh/ForTIFAI.

## Code availability

All code is available at https://github.com/pedramaghazadeh/ForTIFAI.

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## Author contributions

S.Z.S. and P.A. contributed equally to the conceptualization, methodology, and implementation of the study. A.M. provided theoretical insights and manuscript revisions. F.K. supervised the project and provided critical guidance throughout the research. All authors reviewed and approved the final manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

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